Supplementary material S2: per-study quality appraisal

Who Leads, Who Follows? A Review of Leadership and Guidance Dynamics in Evacuation

What this file contains

Every study in the appraised corpus is listed below with its rating on each criterion, so that the criterion-level frequencies reported in Section 3.4 and Figure 5 of the manuscript can be recomputed from the underlying record. The same content is supplied as `appraisal_by_study.csv` for machine reading.

Instruments. Empirical, experimental, and virtual-reality studies were appraised on five criteria adapted from the Mixed Methods Appraisal Tool, version 2018: A1 statement of the research question and appropriateness of the design; A2 description of participants; A3 description of measurements and instruments; A4 completeness of outcome data; A5 treatment of confounders and sources of bias. Simulation studies were appraised on five criteria operationalizing the verification and validation concepts developed for building fire evacuation models: B1 completeness of the model description; B2 justification of parameter values; B3 reporting of verification or validation; B4 reporting of sensitivity or robustness analysis; B5 reproducibility of the simulation setup.

Scale. **R** reported, P partially reported, **N** not reported, – criterion does not apply. No composite score was computed and studies were not ranked; the appraisal characterizes the evidence base rather than grading individual contributions.

Coverage. The table holds 223 studies. The simulation instrument applies to 189 of them and the empirical instrument to 54; 26 studies report both designs and are appraised on both, and 6 report no primary results of their own and are scored on neither.

Table S2.1: Appraisal of each of the 223 studies against the two instruments. R denotes reported, P partially reported, N not reported, and a dash that the criterion does not apply. A1 to A5 are the empirical criteria and B1 to B5 the simulation criteria.

ID	Study	Design	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
101	Ahmed, 2020	empirical	R	R	R	R	P	–	–	–	–	–
102	Albi, 2016	mixed	R	R	R	P	P	R	P	P	P	P
103	Albi, 2020	other	–	–	–	–	–	–	–	–	–	–
104	Albi, 2021	simulation	–	–	–	–	–	R	P	N	P	P
105	Alhindi, 2021	simulation	–	–	–	–	–	R	P	P	P	R
106	Alidmat, 2025	simulation	–	–	–	–	–	R	P	P	R	P
107	Almeida, 2015	simulation	–	–	–	–	–	R	P	N	P	P
108	Almeida, 2021	simulation	–	–	–	–	–	R	P	N	R	P
109	Arteaga, 2023	simulation	–	–	–	–	–	R	R	N	R	P
113	Bellomo, 2016	simulation	–	–	–	–	–	R	P	N	P	P
114	Bode, 2014	empirical	R	R	R	R	R	–	–	–	–	–
115	Bode, 2015	empirical	R	R	R	P	P	–	–	–	–	–
116	Boos, 2014	empirical	R	R	R	R	R	–	–	–	–	–
117	Boukas, 2015	mixed	R	P	R	P	P	R	R	R	P	R
118	Cai, 2014	simulation	–	–	–	–	–	R	P	N	N	P
119	Cao, 2016	simulation	–	–	–	–	–	R	P	N	R	P
120	Cao, 2021	simulation	–	–	–	–	–	R	R	R	N	P

ID	Study	Design	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
121	Cao, 2024	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	P
122	Charalampous, 2014	other	–	–	–	–	–	–	–	–	–	–
123	Chen, 2022	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
124	Chen, 2021	simulation	–	–	–	–	–	**R**	P	**N**	P	P
125	Chen, 2023	simulation	–	–	–	–	–	**R**	P	**R**	**R**	P
126	Cheng, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**N**	P
127	Chennoufi, 2019	simulation	–	–	–	–	–	P	**N**	**N**	**N**	P
128	Chennoufi, 2020	simulation	–	–	–	–	–	**R**	P	P	**R**	**R**
129	Chu, 2019	simulation	–	–	–	–	–	**R**	P	P	**R**	P
131	Cimellaro, 2019	mixed	**R**	**R**	**R**	P	**N**	**R**	**R**	**R**	**R**	**R**
132	*Coordinated Robot-Assisted Human C. . .* , 2018	simulation	–	–	–	–	–	**R**	P	**N**	**N**	P
134	Ding, 2020	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
135	Ding, 2021	mixed	**R**	**R**	**R**	P	P	P	**R**	**R**	**R**	P
137	Ding, 2024	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	**R**
138	Dong, 2014	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
139	Dong, 2022	simulation	–	–	–	–	–	**R**	P	P	**R**	P
140	Dong, 2024	simulation	–	–	–	–	–	**R**	P	P	**R**	P
141	Drury, 2009	empirical	**R**	**R**	**R**	**R**	**R**	–	–	–	–	–
142	Du, 2018	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
143	Elzie, 2023	empirical	**R**	**R**	**R**	**R**	**R**	–	–	–	–	–
144	Fachri, 2017	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
146	Faria, 2010	empirical	**R**	**R**	**R**	**R**	**R**	–	–	–	–	–
147	Feuz, 2013	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
148	Fu, 2021	empirical	**R**	**R**	**R**	**R**	**R**	–	–	–	–	–
149	Fujihara, 2014	simulation	–	–	–	–	–	**R**	P	P	**R**	P
151	Georgoudas, 2007	simulation	–	–	–	–	–	**R**	P	P	**R**	P
152	Gretenkort, 2002	empirical	**R**	P	**R**	P	P	–	–	–	–	–
153	Guan, 2019	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
155	Gul, 2024	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
156	Ha, 2012	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
159	Haghani, 2017	empirical	**R**	**R**	**R**	**R**	**R**	–	–	–	–	–
160	Haghani, 2019	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
163	Han, 2017	simulation	–	–	–	–	–	**R**	P	**N**	P	P
164	Henein, 2010	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
165	Hou, 2014	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
166	Huang, 2022	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
167	Irnich, 2023	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	**R**
168	Ito, 2021	simulation	–	–	–	–	–	**R**	P	**N**	P	P
169	Ivo, 2021	simulation	–	–	–	–	–	**R**	P	**N**	P	P
170	Jensen, 2019	empirical	P	P	**R**	P	P	–	–	–	–	–
171	Ji, 2007	simulation	–	–	–	–	–	**R**	P	**N**	P	P
172	Ji, 2018	mixed	**R**	P	**R**	**R**	P	**R**	**R**	**R**	P	P

ID	Study	Design	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
173	Jiang, 2016	simulation	–	–	–	–	–	**R**	**R**	P	**R**	**R**
175	Jiang, 2020	simulation	–	–	–	–	–	**R**	P	P	**R**	P
176	Jiayang, 2021	mixed	**R**	**R**	**R**	P	P	**R**	**R**	**R**	P	**R**
177	Junaedi, 2013	simulation	–	–	–	–	–	**R**	P	**N**	P	**R**
178	Juniastuti, 2016	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
179	Kakizaki, 2015	mixed	**R**	**R**	**R**	P	P	**R**	P	**R**	P	P
180	Khalid, 2019	simulation	–	–	–	–	–	P	**N**	**N**	**N**	P
181	Khamis, 2020	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
182	Klatt, 2024	other	–	–	–	–	–	–	–	–	–	–
183	Le, 2017	simulation	–	–	–	–	–	**R**	P	**R**	**R**	P
184	Li, 2016	empirical	**R**	**R**	**R**	P	**R**	–	–	–	–	–
185	Li, 2019	simulation	–	–	–	–	–	**R**	P	**N**	P	P
186	Li, 2019	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
187	Li, 2020	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
188	Li, 2020	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
189	Li, 2021	simulation	–	–	–	–	–	**R**	P	**R**	P	P
190	Li, 2021	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	P
191	Li, 2022	simulation	–	–	–	–	–	**R**	P	P	**R**	P
192	Li, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
193	Li, 2024	simulation	–	–	–	–	–	**R**	P	P	**R**	**R**
194	Lima, 2017	other	–	–	–	–	–	**R**	P	**N**	**R**	P
195	Lin, 2018	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
196	Liu, 2018	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	P
197	Liu, 2018	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	P
198	Liu, 2018	simulation	–	–	–	–	–	**R**	P	P	**R**	P
199	Liu, 2018	simulation	–	–	–	–	–	**R**	**R**	**R**	P	P
200	Liu, 2018	simulation	–	–	–	–	–	**R**	P	P	**R**	P
201	Liu, 2018	simulation	–	–	–	–	–	**R**	P	P	**R**	P
202	Liu, 2019	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
204	Liu, 2022	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
205	Lombardi, 2020	empirical	**R**	**R**	**R**	**R**	P	–	–	–	–	–
206	Long, 2020	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
207	Lopez, 2021	simulation	–	–	–	–	–	**R**	P	**N**	**R**	**R**
208	Lopez, 2022	simulation	–	–	–	–	–	**R**	P	**N**	**R**	**R**
209	Lu, 2017	mixed	**R**	**R**	**R**	**R**	P	**R**	**R**	**R**	**R**	P
210	Lu, 2021	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
211	Lu, 2021	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
212	Lu, 2024	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
213	Luh, 2012	simulation	–	–	–	–	–	**R**	P	P	**R**	**R**
214	Luna, 2018	other	–	–	–	–	–	–	–	–	–	–
215	Ma, 2016	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	P
216	Ma, 2017	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
217	Ma, 2017	empirical	**R**	**R**	**R**	**R**	P	–	–	–	–	–

ID	Study	Design	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
218	Ma, 2020	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
219	Makmul, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
220	Mao, 2019	simulation	–	–	–	–	–	**R**	P	**N**	P	P
221	Mao, 2020	simulation	–	–	–	–	–	**R**	P	P	**R**	P
222	Mao, 2024	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
223	Marsh, 2014	mixed	**R**	P	**R**	P	P	P	P	P	**N**	P
224	Mason, 2020	simulation	–	–	–	–	–	**R**	P	**N**	P	P
225	Minton, 2004	simulation	–	–	–	–	–	P	P	**N**	**R**	P
230	Muhammad, 2018	simulation	–	–	–	–	–	**R**	P	**N**	P	P
231	Nayyar, 2021	simulation	–	–	–	–	–	**R**	P	P	**R**	P
232	Nayyar, 2020	empirical	**R**	**R**	**R**	**R**	P	–	–	–	–	–
233	Nayyar, 2023	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
235	Okaya, 2011	simulation	–	–	–	–	–	**R**	P	**N**	P	**R**
236	Pelechano, 2006	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
237	Pinkert, 2007	other	–	–	–	–	–	–	–	–	–	–
238	Pluchino, 2015	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
239	Qiao, 2024	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
240	Qiu, 2010	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
244	Ren, 2009	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
245	Ren, 2021	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
246	Ren, 2022	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
247	Ren, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
248	Rio, 2014	mixed	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	P
250	Robinette, 2012	simulation	–	–	–	–	–	**R**	P	P	**R**	P
253	Sakour, 2016	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
255	Sattaratnamai, 2020	other	–	–	–	–	–	**R**	P	P	P	**R**
256	Shao, 2015	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
257	Sharma, 2009	simulation	–	–	–	–	–	**R**	P	**N**	P	P
258	Sharma, 2011	simulation	–	–	–	–	–	P	**N**	**N**	**N**	P
259	Sharma, 2017	simulation	–	–	–	–	–	P	**N**	**N**	**N**	P
260	Sharpanskykh, 2011	simulation	–	–	–	–	–	**R**	P	P	**R**	P
262	Shendarkar, 2006	mixed	**R**	P	**R**	P	P	**R**	**R**	P	**R**	P
263	Shendarkar, 2008	mixed	**R**	P	**R**	**R**	P	**R**	**R**	P	P	P
264	Shiwakoti, 2014	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
265	Singh, 2009	mixed	**R**	**R**	**R**	P	P	**R**	**R**	**R**	P	P
266	Sirakoulis, 2014	other	–	–	–	–	–	–	–	–	–	–
267	Sirakoulis, 2015	other	–	–	–	–	–	–	–	–	–	–
269	Sobhana, 2023	mixed	**R**	**R**	**R**	P	P	**R**	**R**	**R**	**R**	P
270	Spartalis, 2014	simulation	–	–	–	–	–	**R**	P	**N**	P	P
271	Sriniketh, 2023	mixed	**R**	P	**R**	P	P	**R**	P	**R**	**R**	**R**
272	Srinivasan, 2017	simulation	–	–	–	–	–	**R**	P	**R**	**R**	P
273	Srinivasan, 2018	simulation	–	–	–	–	–	**R**	P	**N**	P	P
274	Sun, 2021	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P

ID	Study	Design	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
275	Taneja, 2018	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	P
276	Tang, 2016	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
277	Tian, 2024	mixed	**R**	P	**R**	P	P	**R**	P	**R**	**R**	P
278	Tianhe, 2018	simulation	–	–	–	–	–	**R**	P	**N**	P	P
279	Tresa, 2021	other	–	–	–	–	–	–	–	–	–	–
280	Trivedi, 2018	simulation	–	–	–	–	–	**R**	P	**N**	P	P
281	Tsorvas, 2016	simulation	–	–	–	–	–	**R**	P	**R**	**R**	P
282	Turgut, 2022	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	**R**
283	Urata, 2012	mixed	**R**	P	**R**	P	P	**R**	**R**	P	P	P
284	Vackova, 2020	empirical	**R**	P	P	P	P	–	–	–	–	–
285	Vani, 2013	simulation	–	–	–	–	–	**R**	**N**	**N**	P	P
287	Voyer, 2016	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
288	Wagner, 2020	other	–	–	–	–	–	–	–	–	–	–
290	Wan, 2020	simulation	–	–	–	–	–	**R**	P	P	**R**	P
291	Wang, 2019	simulation	–	–	–	–	–	**R**	P	**N**	P	P
292	Wang, 2019	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
293	Wang, 2012	simulation	–	–	–	–	–	**R**	P	**N**	P	P
294	Wang, 2015	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
295	Wang, 2017	mixed	**R**	P	**R**	**R**	P	**R**	P	P	P	P
296	Wang, 2017	simulation	–	–	–	–	–	**R**	P	**N**	P	P
297	Wang, 2018	simulation	–	–	–	–	–	**R**	P	P	**R**	P
298	Wang, 2019	simulation	–	–	–	–	–	**R**	P	P	**R**	P
300	Wang, 2022	simulation	–	–	–	–	–	**R**	**N**	**N**	**R**	P
301	Wang, 2022	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
302	Wei, 2011	simulation	–	–	–	–	–	**R**	**N**	**N**	**N**	P
303	Wong, 2017	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
304	Wu, 2014	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
305	Wu, 2022	simulation	–	–	–	–	–	**R**	P	**N**	P	P
306	Wu, 2024	simulation	–	–	–	–	–	**R**	P	**R**	**R**	P
307	Wu, 2024	simulation	–	–	–	–	–	**R**	P	P	**R**	P
308	Xie, 2020	empirical	**R**	P	**R**	P	**R**	–	–	–	–	–
309	Xie, 2021	mixed	**R**	**R**	**R**	P	P	**R**	**R**	**R**	**R**	P
310	Xie, 2022	simulation	–	–	–	–	–	**R**	P	**R**	**R**	P
312	Xie, 2023	simulation	–	–	–	–	–	**R**	P	P	**R**	P
313	Xu, 2021	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
314	Xu, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
315	Xu, 2024	mixed	**R**	P	**R**	P	P	**R**	**R**	**R**	P	P
316	Yang, 2014	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
317	Yang, 2016	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
319	Yang, 2020	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
321	Yi, 2022	mixed	**R**	P	P	P	P	**R**	P	P	**R**	P
322	Yi, 2024	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
324	Yu, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P

ID	Study	Design	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
325	Yu, 2022	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
326	Yu, 2022	mixed	**R**	**R**	**R**	P	P	**R**	**R**	P	P	P
327	Yu, 2024	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
328	Yuan, 2023	simulation	–	–	–	–	–	**R**	P	**R**	P	P
329	Yuecheng, 2013	simulation	–	–	–	–	–	**R**	P	**R**	**N**	P
330	Zell, 2019	empirical	**R**	**R**	**R**	**R**	P	–	–	–	–	–
331	Zhang, 2013	simulation	–	–	–	–	–	**R**	P	**N**	P	P
332	Zhang, 2021	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
333	Zhang, 2018	simulation	–	–	–	–	–	**R**	P	**R**	P	P
334	Zhang, 2021	empirical	**R**	**R**	**R**	**R**	P	–	–	–	–	–
335	Zhang, 2021	mixed	**R**	P	P	P	P	**R**	P	P	P	P
336	Zhang, 2022	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
337	Zhang, 2023	mixed	**R**	P	**R**	P	P	**R**	**R**	**R**	**R**	P
338	Zhang, 2023	simulation	–	–	–	–	–	**R**	P	**N**	**R**	P
339	Zhang, 2023	mixed	**R**	P	**R**	P	P	**R**	**R**	**R**	**R**	P
340	Zhang, 2023	mixed	**R**	P	**R**	P	P	**R**	P	P	P	P
341	Zhang, 2024	simulation	–	–	–	–	–	**R**	P	**N**	**R**	**R**
342	Zhang, 2024	simulation	–	–	–	–	–	**R**	P	P	P	P
343	Zhao, 2018	simulation	–	–	–	–	–	**R**	**R**	**R**	**R**	P
345	Zhao, 2021	simulation	–	–	–	–	–	**R**	P	P	**R**	**R**
346	Zheng, 2019	simulation	–	–	–	–	–	**R**	P	P	P	**R**
347	Zheng, 2020	simulation	–	–	–	–	–	**R**	P	**N**	P	**R**
348	Zheng, 2022	simulation	–	–	–	–	–	**R**	P	P	P	P
349	Zheng, 2024	simulation	–	–	–	–	–	**R**	P	P	**R**	P
350	Zhou, 2018	simulation	–	–	–	–	–	**R**	P	P	**R**	P
352	Zhou, 2019	simulation	–	–	–	–	–	**R**	P	P	**R**	P
353	Zhou, 2022	simulation	–	–	–	–	–	**R**	P	P	P	**R**
354	Zhou, 2022	simulation	–	–	–	–	–	**R**	P	P	P	P
355	Zhou, 2023	simulation	–	–	–	–	–	**R**	P	P	**R**	P
356	Zhu, 2016	empirical	**R**	**R**	**R**	P	P	–	–	–	–	–
357	Zhu, 2021	simulation	–	–	–	–	–	**R**	**R**	P	**R**	P
358	Zou, 2020	simulation	–	–	–	–	–	**R**	**R**	**N**	**R**	**R**
401	Wagner, 2021	other	–	–	–	–	–	–	–	–	–	–
402	Radianti, 2014	other	–	–	–	–	–	–	–	–	–	–
403	Vihas, 2012	simulation	–	–	–	–	–	**R**	P	**N**	**N**	P